

DL and GenAI: 2nd Half Topics Covered

(A) Traditional NLP & Sequence Models

- N-gram language models
- Perplexity
- RNN vs LSTM vs GRU
- Seq2Seq
- Attention mechanism
- Smoothing Techniques

(B) Transformer & Modern LLMs

- Self-attention
- Multi-head attention
- Positional encoding
- Transformer Architecture
- MLM vs autoregressive training
- BERT vs GPT vs T5 vs BART

(C) Prompting & Instruction Tuning

- Zero-shot/few-shot prompting
- Chain-of-thought prompting
- Prompt engineering and prompt tuning
- In-context learning
- Instruction tuning
- Self-Instruct / Evol-Instruct / Orca

(D) RLHF & Alignment

- Reward models
- RLHF pipeline
- Bradley-Terry model
- Importance sampling
- KL regularization
- REINFORCE
- Q-function / Value function / Advantage function
- PPO & PPO-CLIP
- Constitutional AI

NOTE: The above list is only a revision guide and is not exhaustive. Students are strongly advised to study all 10 lecture PDFs thoroughly, including examples, derivations, and discussions covered in the slides. Exam questions may be asked from any content discussed in the lectures.